

Support Vector Machines

Intuition, Geometry, Hard/Soft Margins and Hinge Loss

Sayan CHAKI

LIRIS – Ecole Centrale Lyon – Univ Lyon 2 – INSA

Intuition

A linear classifier may have many separating hyperplanes. SVM chooses the one with the largest safety gap, or *margin*. If perfect separation is impossible, it permits controlled violations through slack variables. Eliminating the slack variables leads directly to hinge loss.

1 Linear classifier and separating hyperplane

For binary labels $y_i \in \{-1, +1\}$ and inputs $x_i \in \mathbb{R}^d$, a linear classifier is

$$f(x) = \text{sign}(w^\top x + b).$$

The decision boundary is $w^\top x + b = 0$. In 2D it is a line, in 3D a plane, and in d dimensions a hyperplane. Points with positive score lie on one side; points with negative score lie on the other.

2 Why not choose any separating hyperplane?

If the classes are separable, several hyperplanes can classify every training point correctly. SVM asks a stronger question: *which separator leaves the largest buffer between the two classes?*

Intuition

Think of the boundary as a wall. A wall placed very close to one class is fragile: a small perturbation can move a point across it. A wall centered in the widest empty corridor is more robust.

3 Distance and the role of the label

The signed distance of x_i from $w^\top x + b = 0$ is

$$\frac{w^\top x_i + b}{\|w\|}.$$

Multiplying by y_i makes the sign meaningful for both classes:

Key equation

$$\text{label-aware signed distance} = \frac{y_i(w^\top x_i + b)}{\|w\|}.$$

For $y_i = +1$, a correct point has a positive classifier score. For $y_i = -1$, a correct point has a negative score, and multiplication by -1 makes the product positive. Thus $y_i(w^\top x_i + b) > 0$ means correct classification.

4 The scaling problem

The parameter pairs (w, b) and $(10w, 10b)$ define the same boundary. Therefore, simply maximizing $y_i(w^\top x_i + b)$ is meaningless: rescaling could make it arbitrarily large without moving the hyperplane. SVM fixes the scale so that the closest training points satisfy

$$\min_i y_i(w^\top x_i + b) = 1.$$

Those closest points are the **support vectors**.

5 Margin boundaries and margin width

After normalization, the two margin hyperplanes are

$$w^\top x + b = 1, \quad w^\top x + b = -1,$$

with the decision boundary $w^\top x + b = 0$ halfway between them. The distance between parallel hyperplanes $w^\top x + b = c_1$ and $w^\top x + b = c_2$ is $|c_1 - c_2|/\|w\|$. Hence

Key equation

$$\text{margin width} = \frac{2}{\|w\|}.$$

The support vectors “support” the solution because moving them can move the optimal boundary, whereas distant points often do not affect it.

6 Hard-margin SVM

Maximizing $2/\|w\|$ is equivalent to minimizing $\|w\|$, and it is convenient to minimize its square:

Key equation

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1 \quad \forall i.$$

For $y_i = +1$, the constraint gives $w^\top x_i + b \geq 1$. For $y_i = -1$, it gives $w^\top x_i + b \leq -1$. Hard margin allows no violations, so it assumes perfectly linearly separable data.

7 Why soft margin?

Real data can contain overlap, noise, outliers and mislabeled observations. Perfect separation may be impossible or undesirable. Introduce one slack variable $\xi_i \geq 0$ per point:

$$y_i(w^\top x_i + b) \geq 1 - \xi_i.$$

Interpretation:

- $\xi_i = 0$: correctly classified and on/outside the margin.
- $0 < \xi_i \leq 1$: correctly classified, but inside the margin.
- $\xi_i > 1$: the signed score is negative, so the point is misclassified.

8 Soft-margin objective and C

The optimization becomes

Key equation

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \end{aligned}$$

The first term prefers a wide margin. The second penalizes violations. A large C makes violations expensive and emphasizes fitting the training data; a small C tolerates more violations in exchange for a wider margin. This is the most useful intuition before discussing bias–variance language.

9 From slack variables to hinge loss

For fixed w, b , the smallest valid slack must obey

$$\xi_i \geq 0, \quad \xi_i \geq 1 - y_i(w^\top x_i + b).$$

Therefore

Key equation

$$\xi_i = \max(0, 1 - y_i(w^\top x_i + b)).$$

Define $z_i = y_i(w^\top x_i + b)$. The hinge loss is

$$L_{\text{hinge}}(z_i) = \max(0, 1 - z_i).$$

If $z_i \geq 1$, the loss is zero. If $0 < z_i < 1$, the point is correct but inside the margin. If $z_i < 0$, it is misclassified.

10 Numerical hinge-loss examples

For $y = +1$:

- If $w^\top x + b = 0.4$, then $z = 0.4$ and $L = 0.6$: correct, but inside the margin.
- If $w^\top x + b = 1.4$, then $L = 0$: safely beyond the margin.
- If $w^\top x + b = -0.5$, then $z = -0.5$ and $L = 1.5$: misclassified.

11 Regularized empirical-risk form

A common equivalent scaling is

Key equation

$$\min_{w,b} \quad \frac{\lambda}{2} \|w\|^2 + \frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i(w^\top x_i + b)).$$

Under the convention in the lecture slide, $\lambda = 1/(nC)$. The precise relationship depends on the chosen scaling convention. Importantly, L_2 regularization has a direct geometric meaning here: smaller $\|w\|$ means larger margin $2/\|w\|$.

12 Hinge loss versus 0–1 and logistic loss

The ideal 0–1 classification loss only asks whether a prediction is right or wrong, but it is discontinuous and difficult to optimize. Hinge loss is a convex surrogate that additionally demands a safety margin. Logistic loss

$$L_{\text{logistic}}(z) = \log(1 + e^{-z})$$

approaches zero as confidence grows but never becomes exactly zero. Hinge loss becomes exactly zero once $z \geq 1$. Logistic regression naturally gives a probabilistic interpretation; the raw SVM score does not directly represent a probability.

13 Why SGD/subgradient methods work

For one point, away from the kink at $z = 1$,

$$\frac{\partial}{\partial w} \max(0, 1 - y(w^\top x + b)) = \begin{cases} -yx, & y(w^\top x + b) < 1, \\ 0, & y(w^\top x + b) > 1. \end{cases}$$

At $z = 1$ the hinge is not differentiable, but it is convex and admits subgradients. A point already safely outside the margin contributes no hinge gradient; a point too close or on the wrong side pushes the model toward a larger signed margin.

14 Teaching the whole story

A useful classroom sequence is:

linear classifier $\rightarrow$ many separators $\rightarrow$ maximum margin $\rightarrow$ hard margin $\rightarrow$ slack $\rightarrow$ hinge loss

Tell students: “There are many lines that separate red and blue points. I do not want an arbitrary line; I want the line with the largest safety buffer.” Then introduce support vectors, allow violations for noisy data, and finally show that the smallest legal slack is exactly hinge loss.

15 Four equations to remember

Decision boundary:	$w^\top x + b = 0,$
Margin width:	$\frac{2}{\ w\ },$
Hard margin:	$\min \frac{1}{2} \ w\ ^2 \text{ s.t. } y_i(w^\top x_i + b) \geq 1,$
Hinge form:	$\min \frac{\lambda}{2} \ w\ ^2 + \frac{1}{n} \sum_i \max(0, 1 - y_i f(x_i)).$

16 Three quantities students often confuse

- **Decision boundary** $w^\top x + b = 0$: where the predicted class changes.
- **Margin boundaries** $w^\top x + b = \pm 1$: desired safety boundaries.
- **Margin score** $y_i(w^\top x_i + b)$: how safely a particular point is classified.

Thus hinge loss is simply $\max(0, 1 - \text{margin score})$.